

Mechanical Digestion

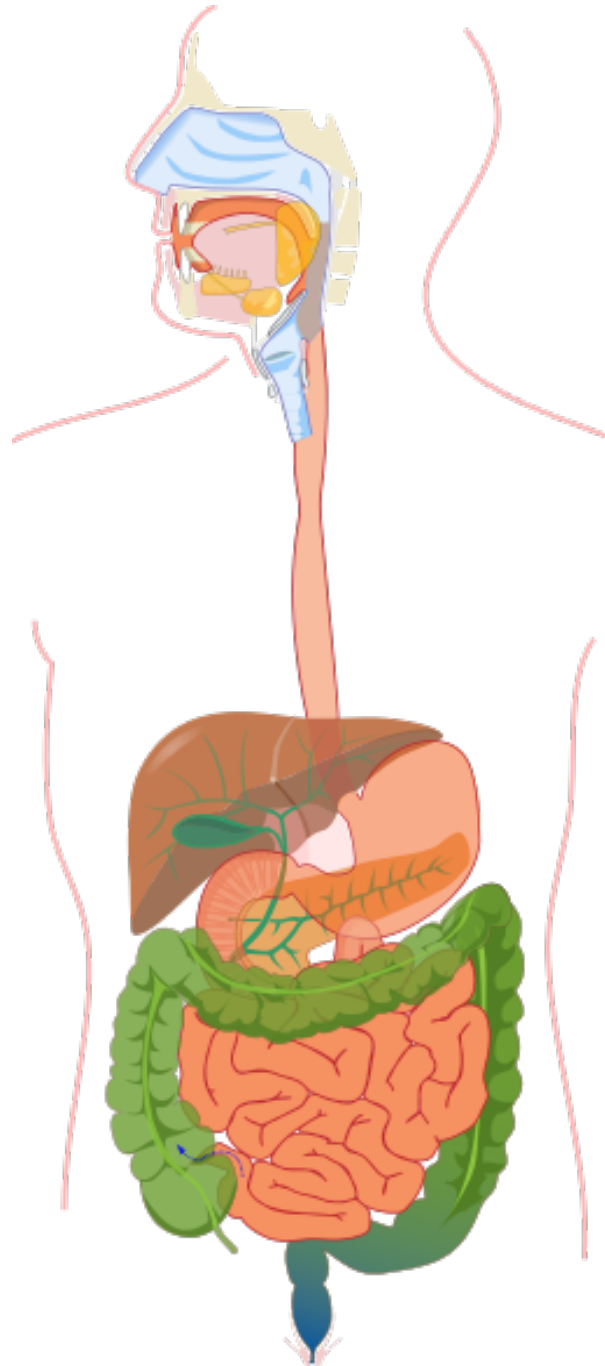
To Do After Lab!

- Keep all the notes and diagrams you create as part of the activities below, as you will be referring to your notes during our lab tests.

Today's Materials for Use at All Stations

(addnl materials per station listed on following pages)

- Colored Pencils
- Blank Pages in Lab Notebook
- Extra Textbooks/Manuals in Room



Station 1

Mouth/Salivary Gland Models and/or Diagrams

Station 1 Mouth Information Sheet

1. Write “Station 1” on the diagram at the start of these activity sheets to indicate where in the digestive system we are at this station.
2. In your notebook, create a rough sketch of a sagittal view of the mouth. Label the oral cavity, tongue, hard palate, soft palate, lingual tonsil, palatine tonsil, and the parotid, sublingual, and submandibular salivary glands.
3. What do you notice about the medical images, if available? Which features of the mouth and salivary glands you labeled above can you see? What, if any, abnormalities might you notice?
4. Review the information at this station regarding oral cancer. Return to your diagram above and use labels and arrows (perhaps in a different color) to indicate the locations of the cancerous cells in the three types of oral cancer discussed.

Station 2

Tooth Models and/or Diagrams

Tooth Microscope Slide

Skull Specimen(s) from Various Species

Station 2 Tooth Information Sheet

1. Write “Station 2” on the diagram at the start of these activity sheets to indicate where in the digestive system we are at this station.
2. In your notebook, create a rough sketch of a typical tooth. Label the crown, neck, root, enamel, dentin, and pulp.
3. Look at the skull(s) at this station. Identify incisors, canines, premolars, and molars. What differences do you notice in the teeth between different species?
4. Look at the slide at this station and record a diagram of your microscope view in your notebook. Label the type of tooth you are looking at, as well as the enamel, dentin, and pulp.
5. Review the information at this station regarding cavities (also known as “caries”). For each of the images, identify and record the type of tooth with the cavities (incisor, canine, premolar, molar), the layers of the tooth impacted by the cavity (enamel, dentin, pulp), and whether the tooth is a deciduous or permanent tooth (to the extent you can tell from the image).

Station 3

Esophagus Models and/or Diagrams

Esophagus Microscope Slide

Medical Image(s) of Esophagus

Station 3 Esophagus Information Sheet

1. Write “Station 3” on the diagram at the start of these activity sheets to indicate where in the digestive system we are at this station.
2. In your notebook, create a rough sketch of the esophagus. Label the pharynx, epiglottis, and gastroesophageal sphincter (also called lower esophageal sphincter).

3. What do you notice about the esophagus medical images, if available? Which features of this organ or structures you labeled above can you see? What, if any, abnormalities might you notice?
4. Look at the slide at this station and record a diagram of your microscope view in your notebook. Look up a diagram of esophagus histology either in a book or online. Label the mucosa, submucosa, and muscularis externa. In addition, look up and label the tissue types in those layers of the esophagus.
5. Review the information at this station regarding gastroesophageal reflux disease. Create a new diagram in your notebook showing a view of the entire length of the esophagus. Use different colors to indicate the mucosa, submucosa, and muscularis externa in the wall (make them sort of thick layers so you can distinguish between them).
6. Label the structure on your diagram most directly responsible for allowing gastric acid to inappropriately enter the esophagus. Also, draw lines to show the axons of pain receptors that will sense pain from the acid. Use the information at this station to show where the pain receptors are located in the esophagus (in which layers/tissues the pain receptor axons are located).

Station 4

Stomach Models and/or Diagrams

Stomach Microscope Slide

Stomach Specimen(s) and/or Medical Images

Station 4 Stomach Information Sheet

1. Write “Station 4” on the diagram at the start of these activity sheets to indicate where in the digestive system we are at this station.
2. In your notebook, create a rough sketch of the stomach. Label the cardia, fundus, body, rugae, and pyloric sphincter.
3. What do you notice about the specimen(s) or medical images, if available? Which features of these organs or structures you labeled above can you see?
4. Look at the slide at this station and record a diagram of your microscope view in your notebook. Look up a diagram of stomach histology either in a book or online. Label the mucosa, muscular layer, and submucosa.
5. Review the information at this station regarding gastric ulcers. Create another diagram of the stomach microscope slide, but this time show what it might look like if an ulcer had caused a perforation. Which layers would the ulcer have had to destroy to create the perforation and in what order?

Station 5

Large Intestine Models and/or Diagrams

Large Intestine Microscope Slide

Large Intestine Specimen(s) and/or Medical Images

Station 6 Large Intestine Information Sheet

1. Write “Station 6” on the diagram at the start of these activity sheets to indicate where in the digestive system we are at this station.

2. In your notebook, create a rough sketch of the large intestine. Label the appendix, cecum, ascending colon, transverse colon, descending colon, rectum, and anal canal.
3. What do you notice about the specimen(s) or medical images, if available? Which features of these organs or structures you labeled above can you see?
4. Look at the slide at this station and record a diagram of your microscope view in your notebook. Look up a diagram of large intestine histology either in a book or online. Label a goblet cell.
5. Review the information at this station regarding colorectal cancer and colonoscopy. Refer to your earlier diagram and list the structures the flexible tube would pass through, from first to last, during a colonoscopy.

Station 6

This station provides a chance to learn about how the organs of our digestive system move materials through the digestive tract. Our digestive tract organs generally include two main smooth muscle layers – a longitudinal layer and a circular layer. These muscle layers work together to perform *peristalsis*, which is a coordinated pattern of contractions that moves materials through the digestive system. We will use some materials in the room to help you demonstrate the activities of these muscles in performing peristalsis

1. Obtain one of the woven finger puzzles and a marble. The finger puzzle will represent the muscular wall of your digestive tract. The marble will represent food.
2. Place the marble inside the tube of the puzzle. Use your fingers to try to simulate the activities of both the longitudinal muscle layer and the circular muscle layer. If done correctly, your efforts should cause the marble to easily pass through the tube.
3. Pass the materials around to the colleagues at your table so everyone can give it a try.
4. Try to figure out...
 - a. What impact does contraction of the longitudinal layer have on the length of the tube?
 - b. What impact does contraction of the longitudinal layer have on the width of the tube?
 - c. Do you think the marble moves most effectively if the longitudinal contractions happen in front of the marble or behind the marble?
 - d. Do you think the marble moves most effectively if the circular contractions happen in front of the marble or behind the marble?
 - e. How would you summarize the main roles of the longitudinal and circular layers?
 - f. What challenges would food encounter in moving through the digestive tract if the longitudinal muscles were not functioning properly?
 - g. What challenges would food encounter in moving through the digestive tract if the circular muscles were not functioning properly?

Reviewing Digestive Organ Fun Facts

Record the answers to the following:

1. Which of the digestive system regions studied today has the strongest muscle in the body (for its size)?
2. Which of the organs studied today includes the hardest substance in the body? And what is that substance?
3. Which of the organs in the digestive tract can be completely removed without major impacts on digestion?
4. In rare instances, swallowing sharp objects, such as fish bones, has resulted in a punctured aorta. Which of the digestive organs studied today would be most responsible for this injury and why?
5. Which of the organs in the digestive tract has an extra, third layer of muscle in its wall?